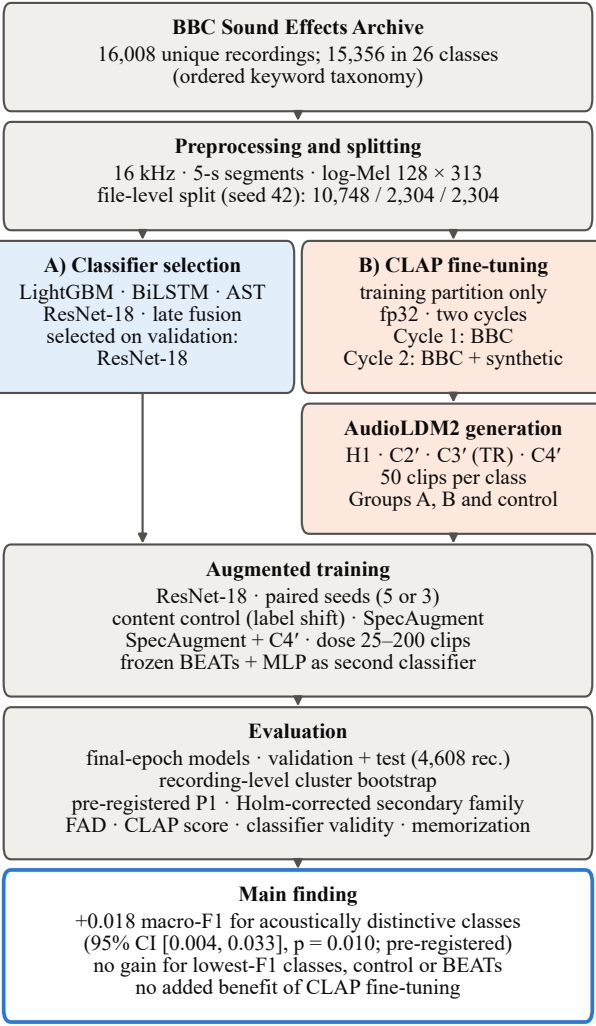

BBC Sound Effects Archive
16,008 unique recordings; 15,356 in 26 classes
(ordered keyword taxonomy)
Preprocessing and splitting
16 kHz · 5-s segments · log-Mel 128 × 313
file-level split (seed 42): 10,748 / 2,304 / 2,304
A) Classifier selection
LightGBM · BiLSTM · AST
ResNet-18 · late fusion
selected on validation:
ResNet-18
B) CLAP fine-tuning
training partition only
fp32 · two cycles
Cycle 1: BBC
Cycle 2: BBC + synthetic
AudioLDM2 generation
H1 · C2′ · C3′ (TR) · C4′
50 clips per class
Groups A, B and control
Augmented training
ResNet-18 · paired seeds (5 or 3)
content control (label shift) · SpecAugment
SpecAugment + C4′ · dose 25–200 clips
frozen BEATs + MLP as second classifier
Evaluation
final-epoch models · validation + test (4,608 rec.)
recording-level cluster bootstrap
pre-registered P1 · Holm-corrected secondary family
FAD · CLAP score · classifier validity · memorization
Main finding
+0.018 macro-F1 for acoustically distinctive classes
(95% CI [0.004, 0.033], p = 0.010; pre-registered)
no gain for lowest-F1 classes, control or BEATs
no added benefit of CLAP fine-tuning